

YEREVAN ZVARTNOTS, Armenia

WMO: 377880

Lat: 40.150N

Lon: 44.383E

Elev: 2802

StdP: 13.27

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	7.4	-2.4	5.3	2.4	3.3	7.2	10.4	17.5	44.7	12.2	42.1	1.0	350	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	98.9	66.2	96.5	66.3	93.4	66.0	71.0	91.6	69.1	89.5	67.5	87.9	5.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	99.1	83.1	62.2	92.7	79.4	60.4	87.0	77.7	36.8	92.1	35.1	89.4	33.7	88.1	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	17.2	13.8	-1.3	103.2	10.8	2.5	-9.0	105.0	-15.3	106.5	-21.3	107.9	-29.2	109.7		
			-1.8	74.1	10.5	2.1	-9.3	75.6	-15.5	76.8	-21.4	78.0	-29.0	79.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	26.0	33.5	45.4	55.4	64.6	74.0	80.6	80.3	70.8	58.0	43.4	31.1	
	DBStd	19.79	9.99	9.15	7.21	6.57	5.32	5.41	5.07	4.81	5.97	5.85	6.55	9.16	
	HDD50	2212	745	463	177	22	0	0	0	0	0	9	210	586	
	HDD65	5010	1209	883	608	295	72	3	0	0	17	224	648	1051	
	CDD50	4172	0	0	34	184	453	720	948	940	623	257	13	0	
	CDD65	1493	0	0	0	6	59	273	483	475	190	7	0	0	
	CDH74	18293	0	0	4	114	670	3145	6036	5986	2198	140	0	0	
	CDH80	8784	0	0	0	23	146	1398	3185	3132	884	16	0	0	
Wind	WSAvg	4.1	1.6	2.3	4.2	4.6	4.5	5.8	7.2	7.0	5.0	3.0	2.2	1.6	
Precipitation	PrecAvg	11.1	0.9	0.9	1.2	1.5	1.7	0.8	0.5	0.3	0.4	1.1	1.0	0.9	
	PrecMax	18.5	2.4	4.0	3.2	4.1	4.5	3.7	1.9	2.0	3.7	3.6	3.0	3.3	
	PrecMin	4.6	0.0	0.1	0.0	0.4	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	PrecStd	3.4	0.7	0.7	0.9	0.9	1.0	0.7	0.4	0.4	0.7	0.9	0.9	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	59.0	72.8	82.8	87.8	98.4	103.5	102.5	96.9	82.4	67.0	53.2	
		MCWB	41.0	45.6	52.1	57.8	60.1	64.5	65.7	66.3	63.8	59.5	52.7	44.6	
	2%	DB	46.4	55.1	66.4	77.0	84.1	94.4	99.6	98.9	91.8	77.4	62.2	48.3	
		MCWB	39.6	43.4	49.6	54.8	59.7	64.0	66.5	66.0	63.6	57.5	49.6	42.0	
	5%	DB	42.6	51.1	62.6	72.6	80.7	91.2	96.6	96.5	88.2	74.8	58.7	45.0	
		MCWB	37.2	41.9	47.4	53.5	58.8	63.7	67.2	66.5	62.7	56.9	48.2	40.4	
	10%	DB	39.1	47.4	58.9	68.8	77.4	87.9	93.4	93.4	85.6	71.4	55.4	42.7	
		MCWB	35.3	40.0	46.2	52.3	57.9	63.3	66.9	66.3	62.2	55.9	46.7	39.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.7	48.0	53.8	59.6	64.2	69.2	73.6	74.0	68.8	62.1	54.5	45.8
			MCDB	47.8	56.1	67.9	75.8	81.0	89.5	94.1	94.3	88.9	75.4	62.6	50.8
2%		WB	40.3	45.0	51.0	56.9	62.2	66.9	71.2	71.3	66.2	59.9	51.7	43.4	
		MCDB	44.2	52.4	62.9	70.3	76.9	87.1	91.9	91.8	86.6	71.8	58.5	46.4	
5%		WB	37.8	42.5	48.9	55.4	60.8	65.6	69.3	69.2	64.4	58.3	49.6	41.3	
		MCDB	41.7	49.5	59.5	67.8	75.6	85.4	90.3	89.1	84.4	70.1	56.3	43.9	
10%		WB	35.2	40.3	47.1	53.8	59.5	64.3	68.0	67.4	62.8	56.9	47.6	39.1	
		MCDB	38.4	46.5	57.2	65.9	73.0	83.4	88.6	87.5	82.1	68.7	54.0	41.9	
Mean Daily Temperature Range	5% DB	MDBR	15.6	19.0	21.6	22.0	23.7	26.4	25.7	26.1	26.9	24.1	21.5	15.2	
		MCDBR	17.7	23.7	26.5	27.8	27.6	29.0	27.5	28.4	30.2	28.8	25.5	18.6	
		MCWBR	11.7	14.1	13.3	12.2	10.4	9.8	9.5	9.8	11.4	13.2	15.1	12.8	
	5% WB	MCDBR	15.4	21.1	22.8	23.7	24.4	27.3	26.1	26.5	28.5	24.5	22.0	15.7	
		MCWBR	10.6	13.0	12.0	11.0	10.4	10.1	10.2	10.7	11.9	11.8	13.8	11.2	
Clear-Sky Solar Irradiance	taub	0.315	0.353	0.413	0.485	0.454	0.432	0.480	0.463	0.396	0.393	0.334	0.311		
	taud	2.312	2.195	2.064	1.917	2.070	2.190	2.064	2.106	2.275	2.210	2.357	2.361		
	Ebn at Noon	266	269	263	252	263	269	254	255	267	251	257	258		
	Edh at Noon	29	37	48	59	52	46	52	48	38	36	27	25		
All-Sky Solar Radiation	RadAvg	738	1064	1259	1474	1810	2199	2205	2022	1649	1081	729	556		
	RadStd	79	134	121	169	105	134	100	101	99	79	97	76		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.01	N/A	N/A	+2.86	-1.67	-2.21	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	+1.48	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page